Experiment No. (Eleven): Electrical conductivity of gases under low pressure

Experimental Theory:

Non-ionized gases are electrically insulating and can be made conductive either by passing an electric current or by cosmic rays through them. This ionization produces an electron and a positive ion. If the electric potential applied to the gas inside the discharge tube is low, the energy acquired by the electron and positive ion will not be sufficient to cause further ionization of the other gas atoms. In this case, the current flowing through the discharge tube is very small. However, if the electric potential applied to the gas inside the discharge tube is high, the electron and positive ion will acquire sufficient kinetic energy to cause further ionization of the remaining atoms through collisions with them. This produces a doubled number of positive ions and negative electrons. The ionization process is repeated continuously, resulting in an electrical discharge. This is accompanied by the emission of light rays from the gas, a rapid increase in the current passing through the tube, and a drop in voltage resulting from the gas inside the tube transforming from a non-conductor to a conductor, and also a decrease in the internal resistance of the gas inside the tube.

The apparatus used in this experiment consists of a long glass tube containing an inert gas under low pressure. A metal electrode (disc-shaped or cylindrical) is attached to each end of the tube.

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Methods :

1. Connect the electrical circuit as shown in the figure, with the ammeter set to measure current with the micrometer, and the voltmeter set to measure voltage up to 6000 volts.

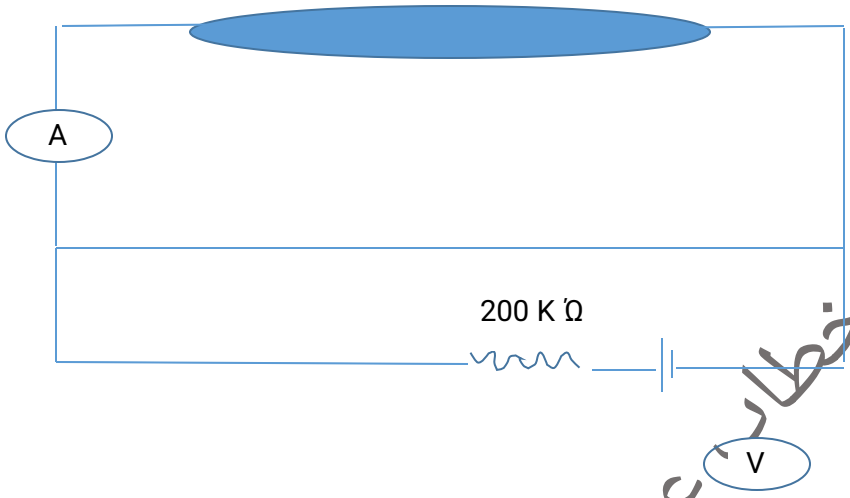
2. Change the voltage from zero to 2500 volts in 500-volt increments, recording the current value in each step.

3. Continue changing the voltage in 100-volt increments, observing the discharge as the voltage increases, while also recording the current value.

4. Change the current scale to milliamperes after the discharge has occurred, and continue changing the voltage while measuring the

current value.

5. Draw the relationship between the current on the vertical axis and the potential difference on the horizontal axis, discuss the curve, and find the discharge voltage.



Discussion Questions:

1. Why is light emitted when an electrical discharge occurs in gases?

22 What is the relationship between free path length and discharge occurrence?

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